

Applying the Oracle, Provenance, and Differentiating the Circuit: Mapping the Geometry and Reading the Filtration

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Stage 11 (parent: doi:local:self_10)

Abstract

Stage 11 of a self-evolving hypergraph, extended from its parent (doi:local:self_10) so the inquiry can look fully back. It declares 8 nodes, each pairing a suggestive Greek glyph with a precise description. Of these, 1 compile via @dslai into small trainable modules (Sigma), making the hypergraph partly differentiable; a module computes, it does not comprehend. It also reads its own published lineage — opening 12 ancestor stages and tracing where each term entered — and records that provenance as data, not as self-understanding. A circuit equation compiles 3 trainable networks — one bound to each symbol — and wires them into 1 composite(s); the equation builds the networks, and the numbers they return are data, not the symbols' meanings combining. It posits 5 fuzzy axioms and 3 large laws over the terms, with memberships taken from the relation geometry; each law evaluates to a number in $[0,1]$ recorded as data — the degree to which an authored postulate holds, not a truth about a self. It inscribes the formal measurements of the companion paper on matrix inscriptions: co-occurrence geometry and Laplacian spectrum, circuit Jacobians as matrix products, and provenance as a filtration with birth times — each a deterministic function of authored data. An oracle increment may extend the hypergraph programmatically; the engine records what was proposed as data beside the questions, not as answers. It relates them by 12 binary relations and 9 hyperedges, and may trigger gated actions whose output is recorded as data, never as an answer. The engine, introspected from its own source, exposes `act_coherence`, `act_neural`, `compile_module`, `forward`, `train`, `act_train`, `act_infer`, `read_lineage`, `lineage_search`, `act_provenance`, `act_circuit`, `act_geometry`, `act_jacobian`, `act_filtration`, `act_spectrum`, `act_oracle`, `act_law`, `parse`, `to_questions`, `title`, `abstract`, `run_actions`, `build_tex`, `build_pdf`, `state_summary`, `build_prompt`, `apply_oracle`, `evolve_auto`, `evolve`. It poses questions and enacts computations beside them; it makes no claim to resolve what a self is.

Symbolic core (inherited relations)

$$\Omega \longrightarrow \Sigma$$

$$\Sigma \models \Sigma$$

$$\Sigma \otimes \Lambda$$

$$\partial \Sigma \stackrel{?}{=} \mu$$

$$\Sigma \not\prec \Omega$$

$$\Phi \longrightarrow \Sigma$$

$$\Phi \rightsquigarrow \Sigma$$

$$T \Rightarrow \Lambda$$

$$\partial \Lambda \rightsquigarrow \mu$$

$$\Delta \dashrightarrow \mu$$

$$\partial T \otimes \Delta$$

$$\Psi \rightsquigarrow \Sigma$$

Circuit (a new equation that wires several networks)

Beyond the inherited relations, this stage adds an equation that binds a small trainable network to *each* symbol and composes them along the arrow:

$$\Sigma = \Phi \longrightarrow \Lambda \longrightarrow \mu \quad (\text{width } 5)$$

Each such equation compiles to 3 trainable networks (one per symbol), wired into a composite. Building the networks is what the equation does; the training losses and outputs the composite returns are recorded as data, not the symbols' meanings combining.

Nodes (suggestive glyph + precise description)

Ω origin — *that from which the self is said to come*

Σ self — *the locus that poses and is altered by the question* $\rightarrow$ module 2->2->2 (sigmoid), trainable

Λ language — *the finite medium any self must pass through*

μ meaning — *what a reader supplies, or finds, in the passage*

Φ other — *the one whose regard first gives the self to itself*

T time — *the direction in which the self cannot return*

Δ difference — *the gap that lets one thing be told from another*

Ψ oracle — *the voice that proposes beside the question without claiming to answer*

Hypergraph (questions, and the data actions returned)

- $\{\Sigma\}$? Can the self reconstruct itself from its own compression? [train]
 $\hookrightarrow$ recorded data (not an answer): {'node': 'Sigma', 'arch': '2->2->2', 'task': 'autoencode',

- 'mse_start': 0.0549, 'mse_final': 0.011, 'weights_from_symbols': True, 'params': 12, 'symbolic_round': 0.000196, 'is': 'an optimization result over a defined toy objective, with the trained weights written to the appendix as glyphs; not the self learning what it is'}
- $\{\Omega, \Sigma, \Lambda\}$? Where, in this inquiry's own history, did origin, self, and language first enter? [provenance]

↪ recorded data (not an answer): {'provenance': {'Omega': {'word': 'origin', 'entered': 'stage 0', 'seen_in': ['0', '1', '10', '11', '2', '3', '4', '5', '6', '7', '8', '9'], 'n_relations': 24}, 'Sigma': {'word': 'self', 'entered': 'stage 0', 'seen_in': ['0', '1', '10', '11', '2', '3', '4', '5', '6', '7', '8', '9'], 'n_relations': 83}, 'Lambda': {'word': 'language', 'entered': 'stage 0', 'seen_in': ['0', '1', '10', '11', '2', '3', '4', '5', '6', '7', '8', '9'], 'n_relations': 32}}, 'is': 'a citation of the lineage; where terms entered, not what they mean'}
 - $\{\Phi, \Lambda, \mu, \Sigma\}$? If a self is wired from other, language, and meaning, what does the composite return? [circuit]

↪ recorded data (not an answer): {'circuit': {'Sigma': {'wiring': 'Phi -> Lambda -> mu', 'networks': {'Phi': {'arch': '5->5->5', 'trained_mse': 0.0683}, 'Lambda': {'arch': '5->5->5', 'trained_mse': 0.057}, 'mu': {'arch': '5->5->5', 'trained_mse': 0.0586}}, 'n_networks': 3, 'composite_output_norm': 1.192}}, 'is': 'several trained networks wired by the equation; the numbers are data, not the symbols' meanings combining'}
 - $\{\Sigma, \Phi, \Lambda, \mu, \Delta, T\}$? To what degree do the authored laws of the self hold under the geometry? [law]

↪ recorded data (not an answer): {'law': {'constitution': 0.49, 'dissolution': 0.6, 'self': 0.1}, 'memberships': {'Omega': 0.42, 'Sigma': 1.0, 'Lambda': 0.58, 'mu': 0.55, 'Phi': 0.49, 'Tau': 0.4, 'Delta': 0.39, 'Psi': 0.1}, 'is': 'the degree to which each authored law holds under degree-derived memberships; numbers in [0,1], not truths about a self'}
 - $\{\Omega, \Sigma, \Lambda, \mu, \Phi, T, \Delta\}$? What is the geometry of the relations — and which term is the hub? [geometry]

↪ recorded data (not an answer): {'degrees': {'Omega': 44, 'Sigma': 105, 'Lambda': 61, 'mu': 58, 'Phi': 51, 'Tau': 42, 'Delta': 41, 'Psi': 10}, 'hub': 'Sigma', 'laplacian_smallest_nonzero': 11.324609, 'laplacian_fiedler': 11.324609, 'is': 'a map of the authored relation structure; hub centrality is a fact about what was written, not a fact about selves'}
 - $\{\Phi, \Lambda, \mu, \Sigma\}$? If the circuit is composed, what is the Jacobian of the composite at a point? [jacobian]

↪ recorded data (not an answer): {'jacobian': {'Sigma': {'wiring': 'Phi -> Lambda -> mu', 'jacobian_norm': 0.0001, 'jacobian_shape': [5, 5], 'is': 'the matrix product of per-module Jacobians; composition of maps, not combination of meanings'}}, 'is': 'chain-rule Jacobians of the circuit composite; data, not semantics'}
 - $\{\Omega, \Phi, T, \Delta\}$? At which stage did each term enter the lineage? [filtration]

↪ recorded data (not an answer): {'filtration': {'Omega': {'word': 'origin', 'birth_time': 0, 'iota': 'stage 0'}, 'Phi': {'word': 'other', 'birth_time': 1, 'iota': 'stage 1'}, 'Tau': {'word': 'time', 'birth_time': 10, 'iota': 'stage 10'}, 'Delta': {'word': 'difference', 'birth_time': 10, 'iota': 'stage 10'}}, 'is': 'birth times along the monotone chain; citations with dates, not meanings'}
 - $\{\Omega, \Sigma, \Lambda, \mu, \Phi, T, \Delta, \Psi\}$? Where does each term sit in the spectral embedding of the relation Laplacian? [spectrum]

↪ recorded data (not an answer): {'spectrum': {'Omega': [0.1222, 0.5497], 'Sigma': [0.1011, 0.2815], 'Lambda': [0.1373, -0.0286], 'mu': [0.138, -0.0463], 'Phi': [0.1256, 0.303], 'Tau': [0.1548, -0.4959], 'Delta': [0.1554, -0.5258], 'Psi': [-0.9344, -0.0376]}, 'fiedler_value': 11.324609, 'is': 'a spectral embedding of the co-occurrence matrix; coordinates are functions'}

- of M, not discoveries about selves'}
- $\{\Psi, \Sigma\}$? What increment did the oracle propose for this stage — and what changed?
[oracle]
 $\hookrightarrow$ recorded data (not an answer): {'oracle': {'new_terms': [], 'new_relations': [], 'new_hyperedges': 0, 'new_actions': [], 'stage': '11', 'summary': '+0 term(s), +0 relation(s), +0 hyperedge(s)'}, 'is': 'a record of the authored increment; the proposal stands beside the question, not in place of an answer'}

Axioms and laws (fuzzy logic over the terms)

Memberships are taken from the relation geometry, $\mu(\sigma) = d(\sigma) / \max_{\tau} d(\tau)$:

$$\mu(\Sigma)=1.00, \quad \mu(\Lambda)=0.58, \quad \mu(\mu)=0.55, \quad \mu(\Phi)=0.49, \quad \mu(\Omega)=0.42, \quad \mu(T)=0.40, \quad \mu(\Delta)=0.39, \quad \mu(\Psi)=0.10$$

Axioms (authored postulates, not verified truths):

- GENESIS: $\Omega \Rightarrow \Sigma$
- ALTERITY: $\Sigma \Rightarrow \Phi$
- MEDIATION: $\Sigma \Rightarrow \Lambda$
- DISTINCTION: $\Sigma \Rightarrow \Delta$
- FINITUDE: $\Sigma \Rightarrow T$

Laws. Each large equation brings several terms together; the braces draw the reading. Each evaluates to a number in $[0, 1]$ — recorded as data, not as a verdict.

$$\begin{array}{c}
 \text{given: by origin and by the other} \\
 \Sigma \Longleftrightarrow \overbrace{\Omega \oplus \Phi} \quad \otimes \quad \overbrace{\Lambda \Rightarrow \mu} \\
 \text{reached: through language, toward meaning} \\
 \\
 \text{difference shatters meaning} \\
 \neg \Sigma \Longleftarrow \overbrace{\Delta \otimes \mu} \quad \oplus \quad \overbrace{T \otimes \Lambda} \\
 \text{time writes upon language} \\
 \\
 \text{the mediated} \\
 \Sigma = \underbrace{\Omega \oplus \Phi}_{\text{the given}} \otimes \overbrace{\Lambda \Rightarrow \mu} \ominus \underbrace{\Delta \otimes T}_{\text{the dissolving}}
 \end{array}$$

Under the memberships above, the laws evaluate to

$$\text{CONSTITUTION} \rightsquigarrow 0.490, \quad \text{DISSOLUTION} \rightsquigarrow 0.600, \quad \text{SELF} \rightsquigarrow 0.100$$

degrees to which authored postulates hold under authored memberships — data, not truths about a self.

Questions this stage poses

- Does Ω give rise to Σ — and what precedes Ω ?
- On what does Σ entailing Σ rest?
- When Σ is carried by Λ , what survives and what is lost?
- Is $\partial \Sigma$ the same as μ , or is μ supplied by the reader?
- If Σ does not reduce to Ω , where is the remainder?

- Does Φ give rise to Σ — and what precedes Φ ?
- Does Φ haunt Σ — present as an absence?
- What residue remains because T leaves a trace upon Λ ?
- Does $\partial\Lambda$ haunt μ — present as an absence?
- What collapses when Δ shatters the structure of μ ?
- When ∂T is carried by Δ , what survives and what is lost?
- Does Ψ propose beside Σ without claiming to answer?
- Can the self reconstruct itself from its own compression?
- Where, in this inquiry's own history, did origin, self, and language first enter?
- If a self is wired from other, language, and meaning, what does the composite return?
- To what degree do the authored laws of the self hold under the geometry?
- What is the geometry of the relations — and which term is the hub?
- If the circuit is composed, what is the Jacobian of the composite at a point?
- At which stage did each term enter the lineage?
- Where does each term sit in the spectral embedding of the relation Laplacian?
- What increment did the oracle propose for this stage — and what changed?

Lineage read (this stage searched its ancestors)

This stage read 12 ancestor stages (0, 1, 10, 11, 2, 3, 4, 5, 6, 7, 8, 9) from their embedded sources, tracing 8 terms and 121 relations across the lineage. For example: Ω (origin, from stage 0), Σ (self, from stage 0), Λ (language, from stage 0), μ (meaning, from stage 0), Φ (other, from stage 1). This is provenance retrieval — where terms entered and what was said — recorded as data, not a claim to understand the lineage.

Manifold (the geometry of the relations)

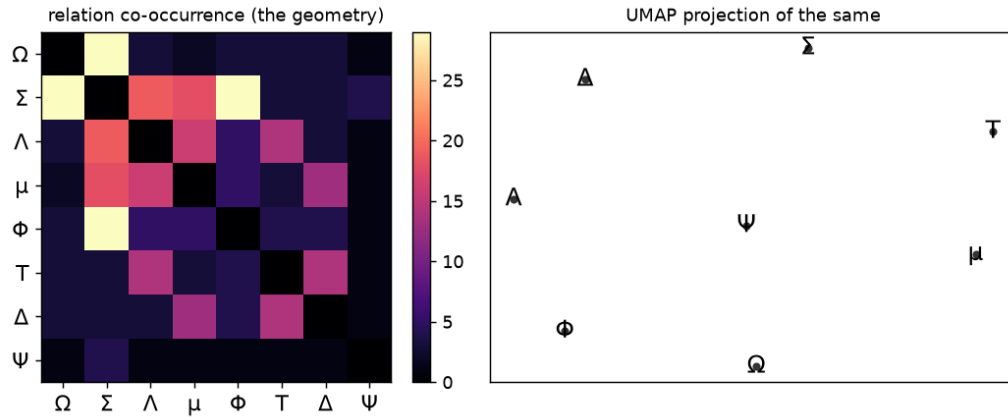


Figure 1: A projection of the relation co-occurrence across the lineage. Left: how often terms share a relation (Σ is densest). Right: a UMAP projection of the same geometry. This is a map of the authored structure — where the terms sit relative to one another — not a discovery of hidden meaning. Computed by an external prototype (`umap_manifold.py`); the engine only embeds it.

Symbolic weights (the trained network, written as glyphs)

Each trained weight is stored as a fraction h/c — an Egyptian hieroglyph over a cuneiform sign — normalized to the recorded range. The runtime reconstructs the network by decoding these glyphs; no binary blob is attached. It looks like esoteric mathematics; it is a quantized weight matrix, and the symbols are the numbers, not meaning. Σ -module $(2 \rightarrow 2 \rightarrow 2)$, 12 weights in range $[-2.45939, 3.523463]$:



Evolution (this stage)

The oracle supplied the increment below; it was applied programmatically and recorded as data beside the questions — not as answers to them.

Oracle (increment):

```
/* introspector7 - stage 11 ; the oracle enters – automated evolution beside the question */
nu : 11 ;
parent : local:self_10 ;

Omega : origin      : "that from which the self is said to come" ;
Sigma : self        : "the locus that poses and is altered by the question" ;
Lambda : language   : "the finite medium any self must pass through" ;
mu      : meaning    : "what a reader supplies, or finds, in the passage" ;
Phi     : other      : "the one whose regard first gives the self to itself" ;
Tau     : time       : "the direction in which the self cannot return" ;
Delta   : difference : "the gap that lets one thing be told from another" ;
Psi     : oracle     : "the voice that proposes beside the question without claiming to answer" ;

module Sigma : 2 -> 2 -> 2 : sigmoid ;
fuzzy : degree ;

Omega -> Sigma ;      /* genesis */
Sigma |= Sigma ;      /* the reflexive claim */
Sigma (x) Lambda ;    /* reached only through language */
@Sigma ?= mu ;         /* is the self's changing the same as meaning? */
Sigma -< Omega ;       /* not merely its origin */
Phi -> Sigma ;         /* the self is given by the other */
Phi ~> Sigma ;         /* the other haunts the self */
Tau *> Lambda ;        /* time writes upon the word */
@Lambda ~> mu ;        /* the shift in language is haunted by vanished meaning */
Delta !> mu ;          /* difference shatters stable meaning */
@Tau (x) Delta ;       /* the passage of time is carried by difference */
Psi <~ Sigma ;         /* the oracle proposes beside the self */

axiom genesis      : Omega => Sigma ;
axiom alterity     : Sigma => Phi ;
axiom mediation    : Sigma => Lambda ;
axiom distinction  : Sigma => Delta ;
axiom finitude     : Sigma => Tau ;
```

```

law constitution : Sigma <=> over( Omega | Phi : "given: by origin and by the other" ) & under( Lambda => mu : "
law dissolution : ~Sigma <= over( Delta & mu : "difference shatters meaning" ) | under( Tau & Lambda : "time wri
law self : Sigma = under( Omega | Phi : "the given" ) & over( Lambda => mu : "the mediated" ) (-) under( Delta &

circuit Sigma = Phi -> Lambda -> mu : 5 ;

{ Sigma } ? "Can the self reconstruct itself from its own compression?" ! train ;
{ Omega, Sigma, Lambda } ? "Where, in this inquiry's own history, did origin, self, and language first enter?" !
{ Phi, Lambda, mu, Sigma } ? "If a self is wired from other, language, and meaning, what does the composite retu
{ Sigma, Phi, Lambda, mu, Delta, Tau } ? "To what degree do the authored laws of the self hold under the geometr
{ Omega, Sigma, Lambda, mu, Phi, Tau, Delta } ? "What is the geometry of the relations – and which term is the h
{ Phi, Lambda, mu, Sigma } ? "If the circuit is composed, what is the Jacobian of the composite at a point?" ! j
{ Omega, Phi, Tau, Delta } ? "At which stage did each term enter the lineage?" ! filtration ;
{ Omega, Sigma, Lambda, mu, Phi, Tau, Delta, Psi } ? "Where does each term sit in the spectral embedding of the
{ Psi, Sigma } ? "What increment did the oracle propose for this stage – and what changed?" ! oracle ;

weights Sigma : 2,2,2 : -1.199002,1.627318 : □□□□□□□□□□□□□□□□□□ ;

```python
stage 11: the propose operator – oracle beside self, not claiming to answer
if "<~" not in OPS:
 OPS.append("<~")
 OP_TEX["<~"] = r"\leadsto"
 OP_Q["<~"] = "Does {a} propose beside {b} without claiming to answer?"
 OP_GLOSS["<~"] = "proposing"
```

```

Application: applied directly to self.dsl

Lineage and provenance

This model extends its parent self-evolving inquiry, doi:local:self_10, ingesting its relations so the series can look fully back [1, 2, 3, 4]. The companion papers set out the method and its discipline [2, 3, 5]. Actions record data beside questions; the document poses and enacts, and makes no claim to resolve what a self is.

Appendix: self-extraction

This PDF carries its engine (introspecter7.py) and DSL (self.dsl) as embedded streams. Requires `pip install pypdf numpy` and a LaTeX install. To produce the next stage, save and run:

```

import base64
exec(base64.b64decode(
    "aW1wb3J0IHN5cywgZ2xvYiwgb3MKc3lzLnBhdGguaw5zZXJ0KDA5IClIuIi"
    "kKZnJvbSBweXBkZiBpbXBvcnQgUGRmUmVhZGVyCmNhbmQgPSB0b25lCmZv"
    "ciBmIGluIGdsb2IuZ2xvYigiKi5wZGYiKToKICAgIHRyeTogYSA9IFBkZl"
    "JlYWRLcihmKS5hdHRhY2htZW50cwogICAgZXhjZXB0IEV4Y2VwdGlvbjog"
    "Y29udGluZWUKICAgIGlmICJpbnRyb3NwZWNOZXI3LnB5IiBpbBiBhIGFuZC"
    "AoY2FuZCBpcyB0b25lIG9yIG9zLnBhdGguZ2V0bXRpbWUoZikgPiBvcy5w"
    "YXRoLmdldG10aW1lKGNhbmQpKToKICAgICAgICBjYw5kID0gZgpwZGYgPS"
    "BjYw5kIG9yIG1heChnbG9iLmdsb2IoIioucGRmIiksIGtleTlvcy5wYXRo"

```

```
"LmdldG10aW1lKQpmb3IgbmFtZSwgZGF0YSBpbjBQZGZSZWFkZXIocGRmKS"  
"5hdHRhY2htZW50cy5pdGVtcygp0gogICAgb3B1bihuYW1lLCAid2IiKS53"  
"cm10ZShkYXRhWzBdIGlmIGlzaW5zdGFuY2UoZGF0YSwgbGlzdCkgZWxzZS"  
"BkYXRhKQppbXBvcnQgaW1wb3J0bGliLCBpbmRyb3NwZWNoZXI3IGFzIEk7"  
"IGltcG9ydGxpYi5yZWxvYWQoSSk7IEkuZXZvbHZlKCKK"))
```

References

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